Analytical Cume

February 10, 2006

This exam covers two papers co-authored by Fran Ligler of the Naval Research Laboratories. These papers were previously posted as this announced cume topic; they are: Anal. Chem. 1993, 65, 3561 and Anal. Chem. 1997, 69, 2779. These will be referred to as the 1993 and 1997 paper respectively.

1. (20 points) The 1993 paper states, "Columns were washed in excess labeled-antigen until a steady baseline was established."

   a. Explain what is happening on the column during this process.

   b. Why is establishing a steady baseline so important at this point in the procedure?

2. (20 points)

   a. Why is more TNB-CV-FL displaced at the lower flow rate?

   b. Why does this leaded to decreased column life?

3. (20 points) Near the end of the paper, Ligler admits that ELISA-methods for TNT achieve lower limits of detection than the continuous-flow method they describe in the paper.

   a. How are ELISA methods able to obtain lower-limits-of-detection?

   b. Even though detection-limits reported in the 1993 paper are not as good as ELISA for TNT, What points does Ligler make about the advantages of the continuous-flow method over ELISA?

4. (40 points) Four years later Ligler in the 1997 paper reports a three-order-of-magnitude improvement in detection limits (20 ppb to 15 ppT, parts per trillion) for TNT. The procedures are basically the same, and some improvements can be attributed to a better antibody-antigen system, and an improved fluorephore.

   a. The majority of the improvement LOD can be attributed to what change in the method between 1993 and 1997?

   b. Give a mechanistic explanation as to why the change generates such an improvement.